



INTRODUCTION TO SYSTEMS, ANALYSIS AND DESIGN

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INTRODUCTION TO SYSTEMS, ANALYSIS AND DESIGN



What is Systems Analysis and Design?

Systems Analysis and Design (SAD) is the process of understanding a system, identifying problems or needs, and designing a better solution. It helps organizations create or improve systems that can support users, processes, information, and business activities.



Examples:



Systems Analysis



Systems Design



Information System

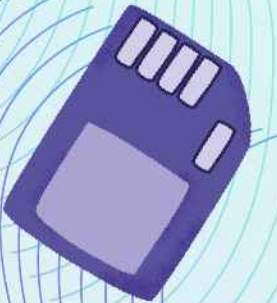




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Information Systems



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Importance of SAD



Analysis, Design, and Implementation



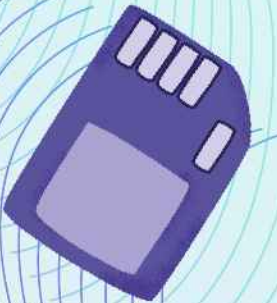
Real-World Example



SAD Tools



Role of a System Analyst





WHAT IS AN INFORMATION SYSTEM?



Information System

An Information System (IS) is a set of components that work together to collect, process, store, and provide information.

Information systems help organizations manage information and support different activities and decisions.



Examples:



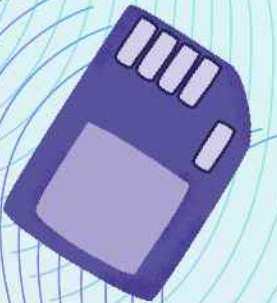
School Information System



Banking System



Ordering System





PURPOSE OF AN INFORMATION SYSTEM



Why Do We Need Information Systems?

Information systems help organizations:



Manage information



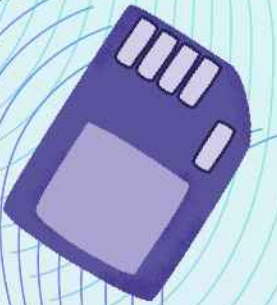
Support decision-making



Coordinate activities



Monitor and analyze information



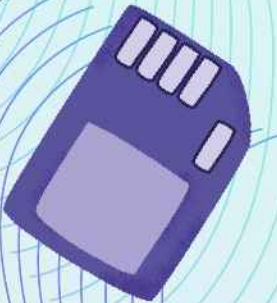


COMPONENTS OF AN INFORMATION SYSTEM

Five Key Components

An information system consists of several important components that work together.

-  Hardware
-  Software
-  Data
-  People
-  Processes





HARDWARE IN AN INFORMATION SYSTEM



Hardware

Hardware refers to the physical equipment used by an information system.

It allows users to enter, process, display, and store information.



Examples:



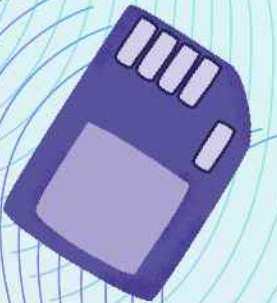
Computer



Keyboard



Printer





SOFTWARE IN AN INFORMATION SYSTEM



Software

Software refers to programs and instructions that tell the computer what to do.

Software helps the information system perform specific tasks.



Examples:



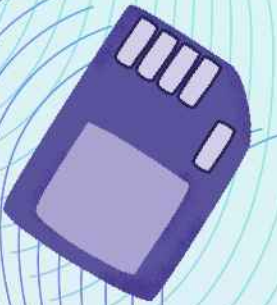
Operating System



Application Software



Database Software





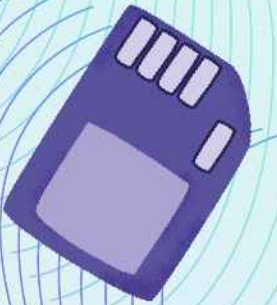
DATA

 **What is Data?**
Data refers to raw facts and details that an information system collects and processes.

Data becomes useful information after it is properly processed.

 **Examples:**

-  **Student Data**
-  **Sales Data**
-  **Customer Data**





PEOPLE

People in an Information System

People are the users and individuals who interact with, manage, or support an information system.



Examples:



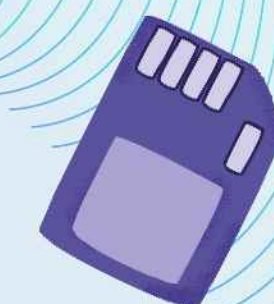
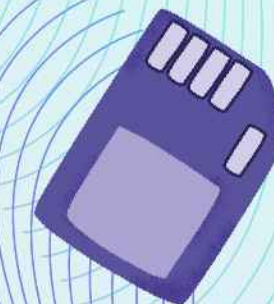
Users



System Analysts



Administrators





PROCESSES



What Are Processes?

A process is a series of steps or procedures followed to complete a task. Information systems often support processes within an organization.



Examples:



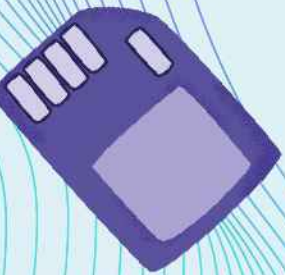
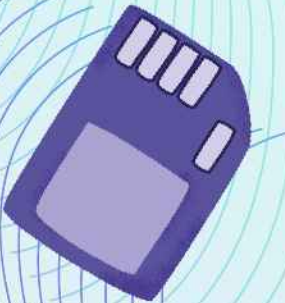
Enrollment Process



Payment Process



Ordering Process

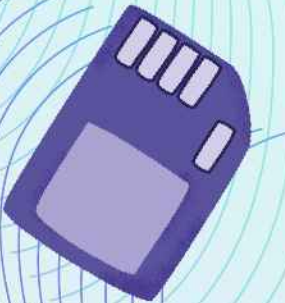




HOW THE FIVE COMPONENTS WORK TOGETHER

 **Components Must Work Together**
Hardware, software, data, people, and processes are connected.
A problem in one component can affect the overall information system.

 **Examples:**
 **Hardware + Software**
 **People + Data**
 **Processes + System**



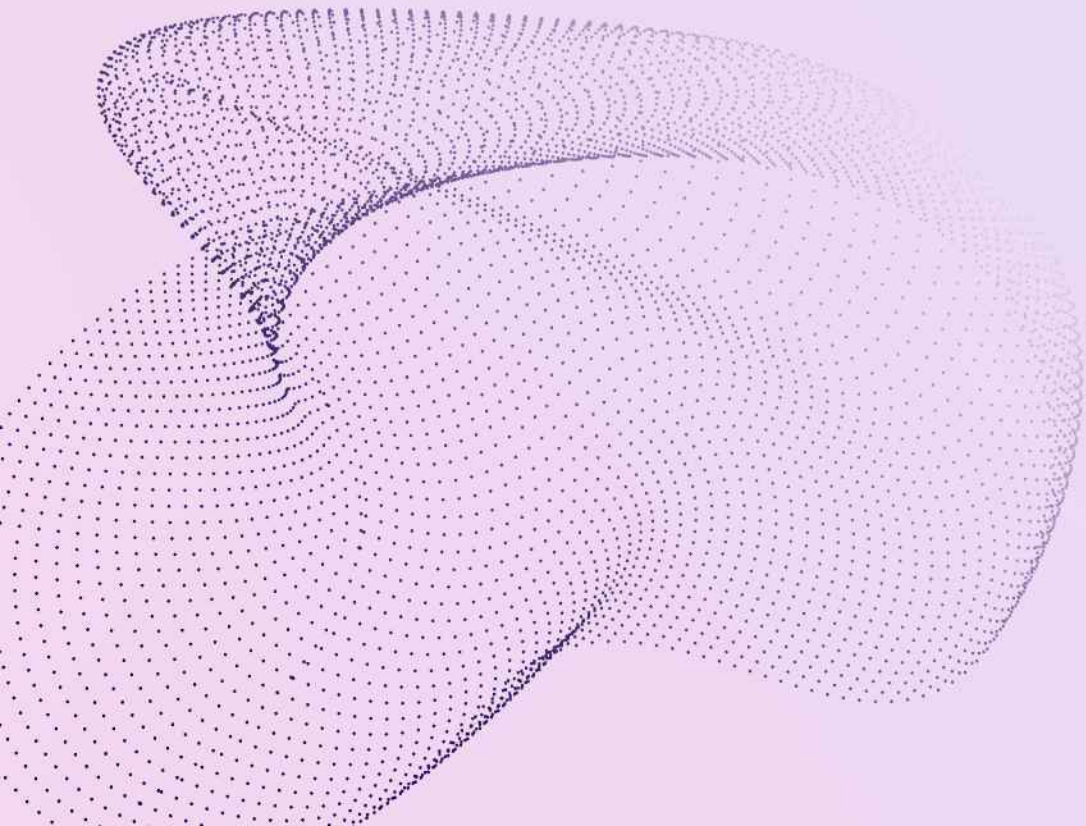


IMPORTANCE OF INFORMATION SYSTEMS

★ Why Are Information Systems Important?

Information systems can help organizations improve:

- Efficiency
- Communication
- Decision-making
- Innovation





EFFICIENCY



Improving Efficiency

Information systems can reduce unnecessary manual work and help tasks become more organized.



Examples:



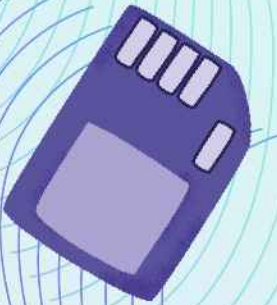
Automation



Faster Processing



Digital Records

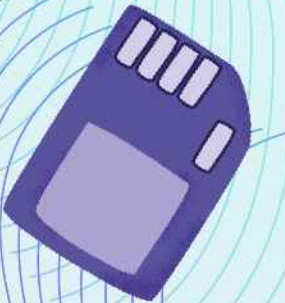




COMMUNICATION

 **Improving Communication**
Information systems can help people share and access information more easily.

 **Examples:**
 **Messages**
 **Notifications**
 **Shared Information**





DECISION-MAKING



Supporting Better Decisions

Information systems provide information that can help users make decisions.



Examples:



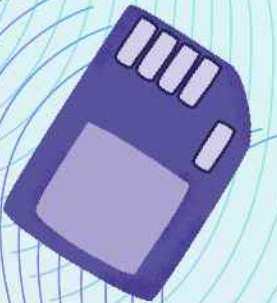
Reports



Statistics



Records





SYSTEMS ANALYSIS

 **What is Systems Analysis?**

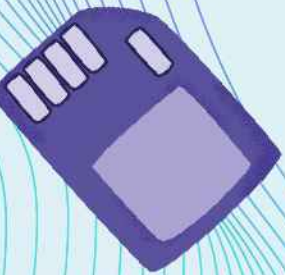
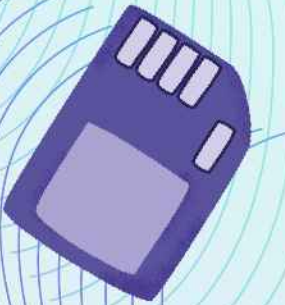
Systems Analysis is the process of studying a system to understand its problems, needs, requirements, and opportunities for improvement.

 **Examples:**

 **Study the Current System**

 **Identify Problems**

 **Find User Needs**





WHY DO SYSTEMS ANALYSIS?



Why Analyze a System?
Systems analysis helps us understand:



What is currently happening?



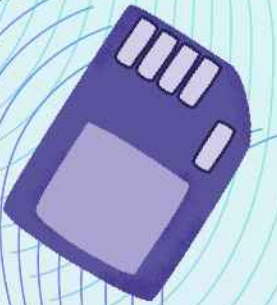
What problems exist?



What do users need?



What can be improved?





SYSTEMS ANALYSIS EXAMPLE



Example: School Enrollment

Imagine a school where students must wait in a long line to enroll. The analyst studies the current enrollment process to identify the cause of the problem.



Examples:



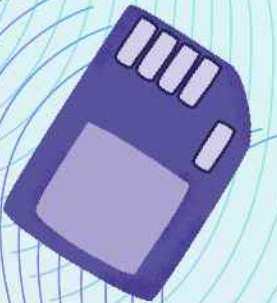
Long Waiting Time



Manual Forms



Repeated Data Entry





USER REQUIREMENTS



What Do Users Need?

User requirements describe what users need the system to do. Understanding these requirements is an important part of systems analysis.



Examples:



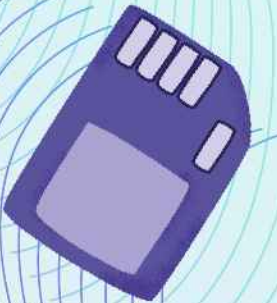
Easy to Use



Fast Access



Accurate Information





SYSTEMS DESIGN



What is Systems Design?

Systems Design is the process of developing and selecting solutions to address the problems and needs identified during analysis.



Examples:



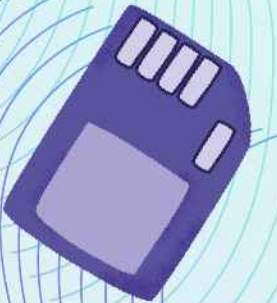
Design Features



Design Processes



Plan the Solution





ANALYSIS VS. DESIGN

➔ Analysis – Understand the problem and requirements.

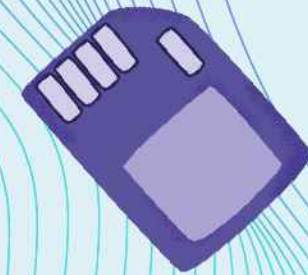
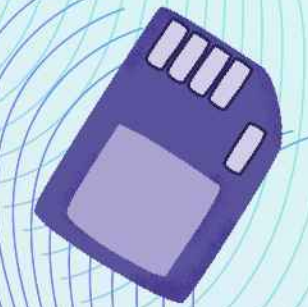
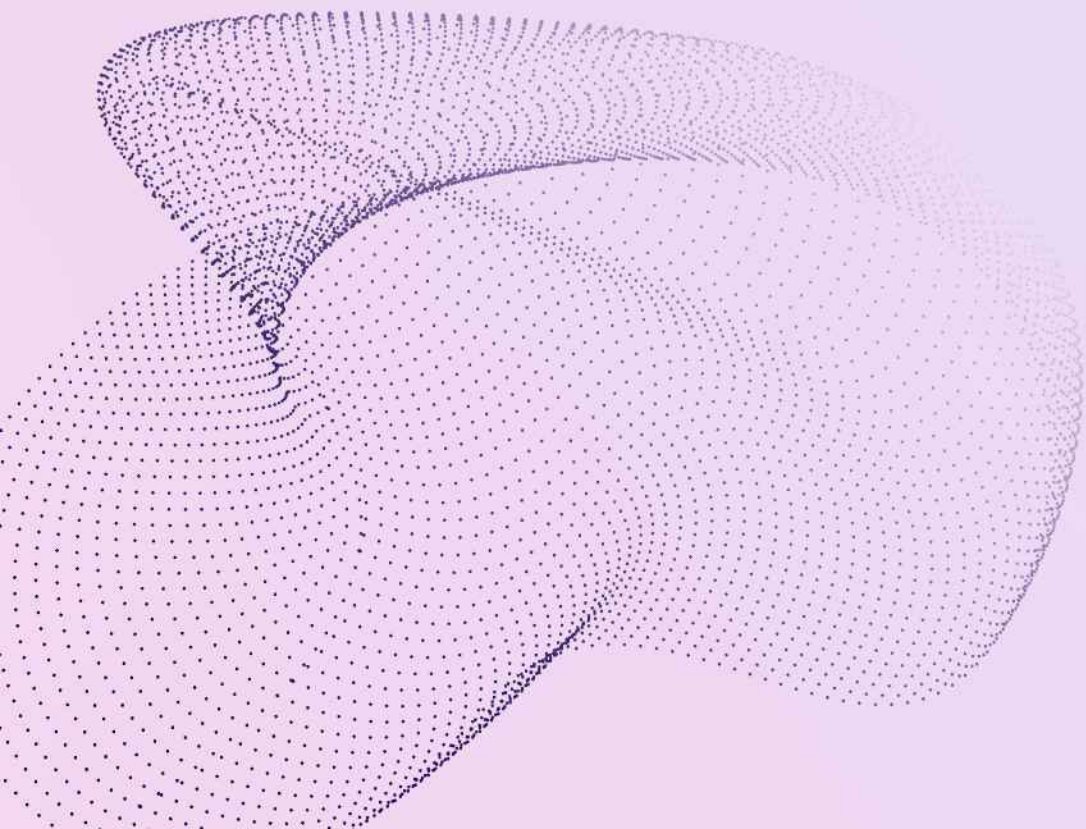
➔ Design – Plan the solution.

Analysis asks:

“What is the problem?”

Design asks:

“How can we solve it?”



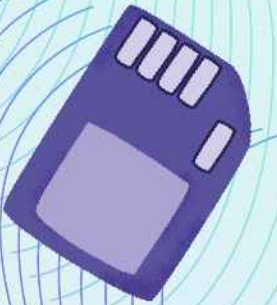


IMPORTANCE OF SYSTEMS ANALYSIS AND DESIGN

★ Why is SAD Important?

Systems Analysis and Design helps organizations:

- Avoid costly mistakes
- Meet user needs
- Provide a system blueprint
- Improve communication





AVOIDING COSTLY MISTAKES



Why Plan Before Building?

Good analysis and design can help identify problems before development becomes expensive.



Examples:



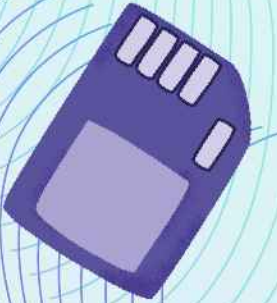
Wrong Requirements



Unnecessary Features



Poor Design





MEETING USER NEEDS



Building for the Users

A system should be designed based on the needs and requirements of its users.



Examples:



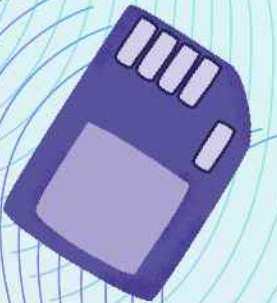
User-Friendly Interface



Useful Features



Relevant Information





SYSTEM BLUEPRINT



What is a System Blueprint?

A system blueprint is a plan or design that shows how the proposed system should work.

It guides developers and other stakeholders during development.



Examples:



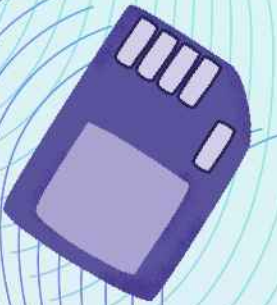
System Structure



Process Flow



Interface Design





ANALYSIS, DESIGN, AND IMPLEMENTATION



From Problem to Working System
The relationship can be simplified as:

ANALYSIS → DESIGN → IMPLEMENTATION



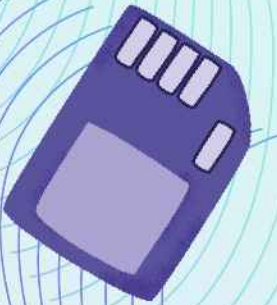
Analysis – WHAT is the problem?



Design – HOW will we solve it?



Implementation – Build and put the solution into use





REAL-WORLD EXAMPLE: GRABFOOD



Example of Systems Analysis and Design

Imagine a food delivery system where customers experience delayed orders.

The system analyst studies the problem and identifies possible improvements such as order tracking and estimated delivery time.



Examples:



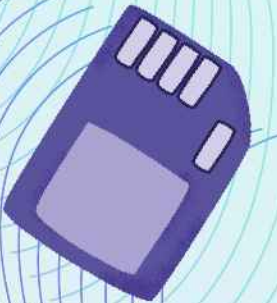
Delayed Orders



Order Tracking



Estimated Delivery Time





SYSTEM ANALYSIS IN THE REAL WORLD



What Does the Analyst Ask?

When analyzing a system, the analyst needs to understand:



What is happening now?



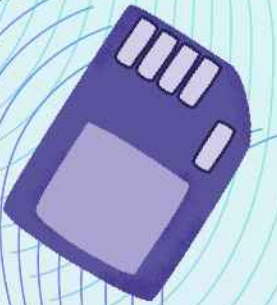
What problem are users experiencing?



Why is the problem happening?



What do users need?





TOOLS USED IN SYSTEMS ANALYSIS AND DESIGN



Common SAD Tools

System analysts use different tools and techniques to understand systems and communicate requirements.



Data Flow Diagram (DFD)



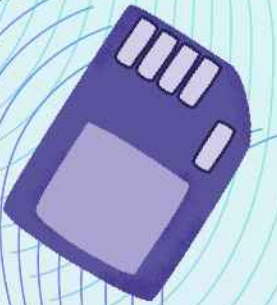
Interviews



Surveys



Documentation





DATA FLOW DIAGRAM



What is a DFD?

DFD means Data Flow Diagram.

It is a visual representation used to show how data moves through a system.



Examples:



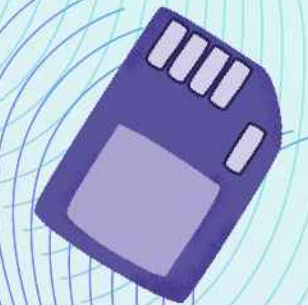
Input Data



Process



Output Data





INTERVIEWS



What is an Interview?

An interview is a technique where the system analyst asks users or stakeholders questions to gather information about the system.



Examples:



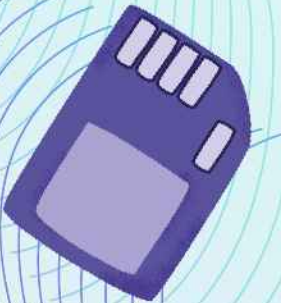
Users



Managers



Employees





SURVEYS



What is a Survey?

A survey collects information from multiple people through a set of questions.

It can help analysts identify common opinions, problems, and preferences.



Examples:



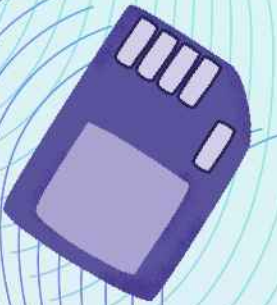
User Satisfaction



Common Problems



User Preferences





DOCUMENTATION



What is Documentation?

Documentation is the written record of important information about a system.

It helps people understand, develop, maintain, and communicate information about the system.



Examples:



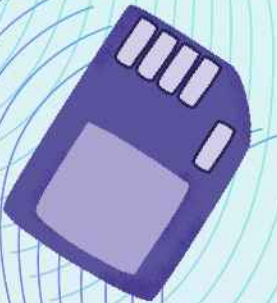
Requirements



Procedures



System Description





ROLE OF A SYSTEM ANALYST



Who is a System Analyst?

A System Analyst studies systems, identifies problems, gathers requirements, and helps design solutions.

The analyst connects the needs of users with the proposed system solution.



Examples:



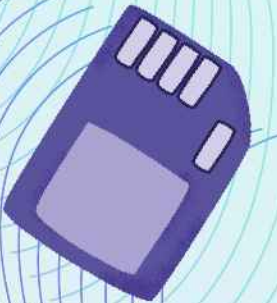
Analyze Problems



Gather Requirements



Communicate with Users





SYSTEM ANALYST AND USERS



Working With Users

System analysts need to communicate with users and stakeholders to understand the real needs of the system.



Examples:



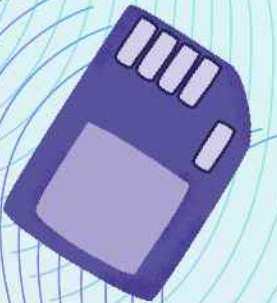
Ask Questions



Listen to Problems



Gather Feedback





SYSTEM ANALYST ACTIVITY



Think Like a System Analyst

Imagine that you are a system analyst.

Choose an existing application and identify three features that you would improve or add.



Examples:



Facebook



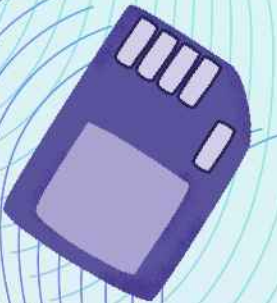
Shopee



Instagram



TikTok





FROM PROBLEM TO SOLUTION



Applying SAD

A good system solution should be based on a clearly identified problem.

Problem → Analysis → Design → Solution



Examples:



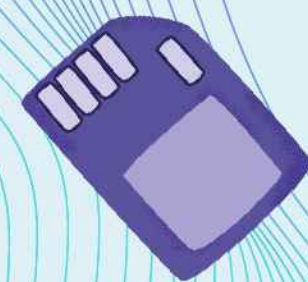
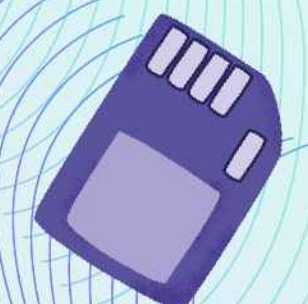
Problem



Analysis



Solution



**THANK
YOU**

